



# Modelling and analysis of the spread of AIDS epidemic with immigration of HIV infectives

Ram Naresh\*, Agraj Tripathi, Dileep Sharma

Department of Mathematics, Harcourt Butler Technological Institute, Kanpur-208 002, India

## ARTICLE INFO

### Article history:

Received 27 July 2007

Received in revised form 20 September 2008

Accepted 25 September 2008

### Keywords:

HIV/AIDS

Infectives

Immigration

Reproduction number

Region of attraction

Stability

## ABSTRACT

We propose and analyze, a nonlinear mathematical model of the spread of HIV/AIDS in a population of varying size with immigration of infectives. It is assumed that susceptibles become infected via sexual contacts with infectives (also assumed to be infectious) and all infectives ultimately develop AIDS. The model is studied using stability theory of differential equations and computer simulation. Model dynamics is also discussed under two particular cases when there is no direct inflow of infectives. On analyzing these situations, it is found that the disease is always persistent if the direct immigration of infectives is allowed in the community. Further, in the absence of inflow of infectives, the endemicity of the disease is found to be higher if pre-AIDS individuals also interact sexually in comparison to the case when they do not take part in sexual interactions. Thus, if the direct immigration of infectives is restricted, the spread of infection can be slowed down. A numerical study of the model is also carried out to investigate the influence of certain key parameters on the spread of the disease.

© 2008 Elsevier Ltd. All rights reserved.

## 1. Introduction

It is well known that immigrants play a critical role in disease dynamics [1,2]. Epidemics ignited or enhanced by immigration of infectious cases include HIV, SARS, avian influenza and measles [3]. Sexually transmitted diseases, such as gonorrhea do not confer immunity, while HIV/AIDS is not yet fully curable. In the proposed model, protective/control measures such as use of condoms, education, complacency, gradual behavioral change are not accounted for. The reason is basically that in rural communities (in the developing world such as Sub-Saharan Africa), individuals are either too poor to afford condoms (partially effective vaccine), and sex education is still a taboo [4]. The health and socioeconomic risks posed by diseases such as HIV/AIDS are compelling scientists to design and implement more effective control and preparedness for emerging and re-emerging infectious diseases. From the initial models of May and Anderson [5–7] various refinements have been added into modelling frameworks, and specific issues have been addressed by researchers, (see for instance, [8–18]). In particular, Busenberg et al. [9] presented a model for HIV/AIDS epidemic growth in Asia by assuming that the spread of disease in the community is mainly caused by the sexual interaction between a core group of female prostitutes and young unmarried males. They obtained several threshold parameters to determine persistence of endemic proportions, persistence of total population and the persistence of infective population. Doyle et al. [12] developed a model for the spread of HIV in a heterosexual population taking into account of the group contact tracing and carried out equilibrium analysis. Greenhalgh et al. [15] studied the impact of condom use on the sexual transmission of HIV and AIDS amongst a homogeneously mixing male homosexual population. Arazoza and Lounes [11] proposed a nonlinear model for an epidemic with contact tracing and applied it to the Cuban HIV/AIDS epidemic to obtain the size of HIV epidemic. Aggarwala [19] analyzed an HIV/AIDS model

\* Corresponding author.

E-mail addresses: [ramntripathi@yahoo.com](mailto:ramntripathi@yahoo.com) (R. Naresh), [agraj\\_hbti@yahoo.co.in](mailto:agraj_hbti@yahoo.co.in) (A. Tripathi), [dilip\\_hbti@yahoo.com](mailto:dilip_hbti@yahoo.com) (D. Sharma).

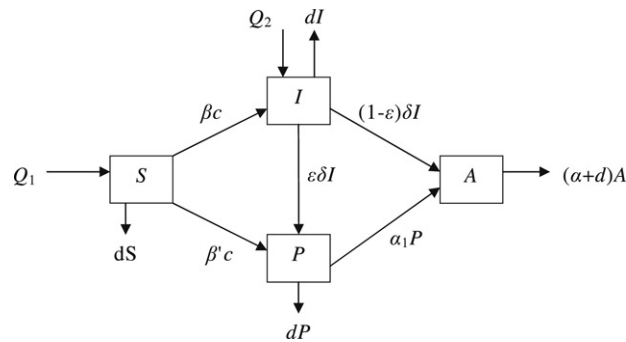


Fig. 1. Transfer diagram of the model.

with a high risk and low risk population and concluded that in Canada, people at high risk are more likely to contact the disease than those who are at low risk. Srinivasa Rao [20] presented a theoretical framework for transmission of HIV/AIDS epidemic in India.

Very often the population under consideration is assumed to be of fixed size with births balancing deaths or migration. This approximation is reasonable when modelling endemic diseases over many years and there are no disease-induced deaths. Since HIV infection and subsequent development of AIDS have led to increased death rates in various risk groups, the demographic structures of the population should be taken into account in modelling of its transmission dynamics [8,10,21].

In the above studies, the direct recruitment of infectives has not been taken into account though it affects the dynamics of the disease to a significant level. Many persons including truck drivers who frequently visit the commercial sex workers get infected and then spread the infection in the community. It is, therefore, essential to consider the direct inflow of infectives to understand the dynamics of the spread of the disease. A very little attention has been paid to study the effect of direct inflow of infectives on the spread of a disease but not implemented in the models of HIV/AIDS [22,1,2]. Brauer and van den Driessche [22] developed a general epidemic model with the assumption that all the immigrants are not susceptible. Kribs-Zaleta et al. [23] developed a model for long distance truck drivers and found that they are at an increased risk of infection due to their frequent contacts with commercial sex workers. Hsieh and Chen [24] developed a theoretical model for a community which has the structure of two classes of commercial sex workers and two classes of sexually active male customers with different levels of sexual activity.

Since susceptibles become infected via sexual contacts with infectives, the infectives then proceed to develop end stage disease AIDS. But due to long incubation period of the disease in adults, it is possible that a patient infected with HIV may develop different epidemiological or clinical stages before developing full blown AIDS depending on the level of infection in an individual, [25,26].

In view of the above, in this paper, a nonlinear mathematical model is proposed to study the transmission of HIV in a varying size population with constant inflow of HIV infectives. Two particular cases of the proposed model are also discussed when there is no direct inflow of infectives. A numerical study of the model is also performed to see the role of various key parameters on the dynamics of the disease.

## 2. Mathematical model

Consider a population of size  $N(t)$  at time  $t$  with constant inflow of susceptibles at a rate  $Q_1$  and that of HIV infectives at a rate  $Q_2$  into the population. The population size  $N(t)$  is divided into four subclasses of susceptibles  $S(t)$ , infectives  $I(t)$  (also assumed to be infectious), pre-AIDS patients  $P(t)$  and that of AIDS patients  $A(t)$  with natural mortality rate  $d$  in all the classes. The susceptibles become infected via sexual contacts with infectives and with those in pre-AIDS class. The total population  $N$  loses individuals at a higher rate from the class  $A$  than the others. It may be noted that the individuals in pre-AIDS class may also interact sexually owing to illiteracy, ignorance or other social factors especially in underdeveloped nations but the contact rate may be very less in comparison to that of other infectives ( $\beta' \ll \beta$ ). It is also assumed that a fraction  $\varepsilon$  ( $0 \leq \varepsilon < 1$ ) of all infectives i.e.  $\varepsilon\delta I$  goes to pre-AIDS class (depending on the level of viral count) while the others with serious infection i.e.  $(1-\varepsilon)\delta I$  directly join the AIDS class. However, it is assumed that virtually all individuals in pre-AIDS class will ultimately develop the disease to join AIDS class. Here  $\delta$  is the rate of movement from infectious class, so that  $(1/\delta)$  denotes the average incubation period,  $\beta$  and  $\beta'$  are the contact rates of susceptibles with infectives and pre-AIDS individuals respectively. The transfer diagram of the model is shown in Fig. 1.

With the above assumptions and considerations, the dynamics of the disease is assumed to be governed by the following system of nonlinear ordinary differential equations,

$$\frac{dS}{dt} = Q_1 - \left( \frac{\beta c S I}{N} + \frac{\beta' c S P}{N} \right) - dS, \quad (2.1)$$

$$\frac{dI}{dt} = Q_2 + \frac{\beta c SI}{N} + \frac{\beta' c SP}{N} - (\delta + d) I, \quad (2.2)$$

$$\frac{dP}{dt} = \varepsilon \delta I - (\alpha_1 + d) P, \quad (2.3)$$

$$\frac{dA}{dt} = (1 - \varepsilon) \delta I + \alpha_1 P - (\alpha + d) A, \quad (2.4)$$

$$S(0) = S_0 > 0, \quad I(0) = I_0 \geq 0, \quad P(0) = P_0 \geq 0, \quad A(0) = A_0 \geq 0$$

where  $\alpha_1$  is the rate by which individuals from pre-AIDS class join AIDS class, the coefficient  $\alpha$  denotes the constant disease-induced death rate and  $c$  is the average number of sexual partners in a unit time.

The above equations can now be written as, using  $N = S + I + P + A$ ,

$$\frac{dN}{dt} = Q_1 + Q_2 - dN - \alpha A \quad (2.5)$$

$$\frac{dI}{dt} = Q_2 + \frac{\beta c(N - I - P - A)I}{N} + \frac{\beta' c(N - I - P - A)P}{N} - (\delta + d) I \quad (2.6)$$

$$\frac{dP}{dt} = \varepsilon \delta I - (\alpha_1 + d) P \quad (2.7)$$

$$\frac{dA}{dt} = \alpha_1 P + (1 - \varepsilon) \delta I - (\alpha + d) A \quad (2.8)$$

$$N(0) = N_0 > 0, \quad I(0) = I_0 \geq 0, \quad P(0) = P_0 \geq 0, \quad A(0) = A_0 \geq 0.$$

Continuity of right-hand side of the system (2.5)–(2.8) and its derivative imply that the model is well posed for  $N > 0$ . It is assumed that all dependent variables and parameters of the model are non-negative.

Now we analyze the model system (2.5)–(2.8) using stability theory of differential equations.

### 3. Equilibrium and stability analysis

In this section, we analyse the existence and stability of the equilibrium points of the model system (2.5)–(2.8).

#### 3.1. Equilibrium of the model

The model system (2.5)–(2.8) does not exhibit a disease-free equilibrium due to direct inflow of infectives at a constant rate. However, there exists only one non-negative equilibrium point of the model (2.5)–(2.8). This endemic equilibrium  $E^* = (N^*, I^*, P^*, A^*)$  exists when HIV infection persists in the population i.e.  $I \neq 0$ , where  $N^*$ ,  $I^*$ ,  $P^*$  and  $A^*$  are positive solutions of the following system of algebraic equations,

$$Q_1 + Q_2 - dN - \alpha A = 0 \quad (3.1)$$

$$Q_2 + \frac{\beta c(N - I - P - A)I}{N} + \frac{\beta' c(N - I - P - A)P}{N} - (\delta + d) I = 0 \quad (3.2)$$

$$\varepsilon \delta I - (\alpha_1 + d) P = 0 \quad (3.3)$$

$$\alpha_1 P + (1 - \varepsilon) \delta I - (\alpha + d) A = 0. \quad (3.4)$$

On solving simultaneous algebraic equations (3.1)–(3.4), we get,

$$I = \tau \pi \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right) \quad (3.5)$$

$$P = \pi \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right) \quad (3.6)$$

$$A = \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right) \quad (3.7)$$

which are all positive when  $N < (Q_1 + Q_2)/d$ .

Here

$$\tau = \frac{(\alpha_1 + d)}{\varepsilon \delta}$$

$$\pi = \frac{(\alpha + d)}{\left[ \alpha_1 + \frac{(1-\varepsilon)}{\varepsilon} (\alpha_1 + d) \right]}$$

and

$$Q_2 N + \pi [\beta c \tau + \beta' c] \left[ N - (1 + \pi + \pi \tau) \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right) \right] \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right) - (\delta + d) \pi \tau N \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right) = 0. \quad (3.8)$$

To show the existence of  $E^*$ , from Eq. (3.8), we write,

$$F(N) = Q_2 N + \pi [\beta c \tau + \beta' c] \left[ N - (1 + \pi + \pi \tau) \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right) \right] \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right) - (\delta + d) \pi \tau N \left( \frac{Q_1 + Q_2 - dN}{\alpha} \right). \quad (3.9)$$

It would be sufficient if we show that the quadratic equation  $F(N) = 0$  has one and only one positive root between 0 and  $(Q_1 + Q_2)/d$ .

To prove this from Eq. (3.9) we have,

$$F(0) = - \left( \frac{Q_1 + Q_2}{\alpha} \right)^2 [\beta c \tau + \beta' c] \pi (1 + \pi + \pi \tau) < 0 \quad (3.10)$$

$$F \left( \frac{Q_1 + Q_2}{d} \right) = Q_2 \left( \frac{Q_1 + Q_2}{d} \right) > 0. \quad (3.11)$$

Also

$$F'(N) = Q_2 + \left[ \{\beta c - (\delta + d)\} \tau \pi + \beta' c + \frac{2\pi d}{\alpha} [\beta c \tau + \beta' c] (1 + \tau \pi + \pi) \right] \left[ \frac{Q_1 + Q_2 - dN}{\alpha} \right] - [\beta c - (\delta + d)\} \tau \pi + \beta' c] \frac{dN}{\alpha}. \quad (3.12)$$

It is easy to note from a theorem in calculus that if  $F'(N) > 0$  then  $F(N) = 0$  has exactly one root (say  $N^*$ ) between 0 and  $\frac{Q_1 + Q_2}{d}$ . Using  $N^*$ , the values of  $I^*$ ,  $P^*$  and  $A^*$  can be found from Eqs. (3.5)–(3.7) respectively.

It is observed from the system (2.5)–(2.8) that if immigration of infectives tends to zero i.e.  $Q_2$  tends to zero then the endemicity of the infection is reduced. This can simply be shown by comparing  $N_{Q_2=0}$  and  $N_{Q_2 \neq 0}$  using a crude assumption,

( $H_1$ ) Let only  $A(t)$  be zero at equilibrium in Eq. (2.5), then,  $\frac{N_{Q_2 \neq 0}}{N_{Q_2=0}} = 1 + \frac{Q_2}{Q_1} > 1$ , this implies that the introduction of infective immigrants increases the population of infectives  $I(t)$ .

### 3.2. Local stability of the equilibrium point

To determine the local stability of  $E^*$ , the following variational matrix is computed about  $E^*$ ,

$$M(E^*) = \begin{bmatrix} -d & 0 & 0 & -\alpha \\ m_{21} & -m_{22} & m_{23} & -m_{24} \\ 0 & \varepsilon \delta & -(\alpha_1 + d) & 0 \\ 0 & (1 - \varepsilon) \delta & \alpha_1 & -(\alpha + d) \end{bmatrix} \quad (3.13)$$

where  $m_{21} = \frac{Q_2}{N^*} + [\beta c - (\delta + d)] \frac{I^*}{N^*} + \frac{\beta' c P^*}{N^*}$ . A sufficient condition for  $m_{21} > 0$  is  $\beta c \geq (\delta + d)$ .

$$m_{22} = \frac{Q_2}{I^*} + \frac{\beta c I^*}{N^*} + \frac{\beta' c P^*}{N^*} + \frac{\beta' c (N^* - I^* - P^* - A^*)}{N^*} \left( \frac{\varepsilon \delta}{\alpha_1 + d} \right) \\ m_{23} = \frac{\beta' c (I^* + P^* + A^*)}{N^*} + \frac{\beta c I^*}{N^*} + \frac{\beta' c P^*}{N^*} - \beta' c \quad \text{and} \quad m_{24} = \frac{\beta c I^*}{N^*} + \frac{\beta' c P^*}{N^*}.$$

The characteristic equation corresponding to  $M(E^*)$  is given by

$$f(\lambda) = \lambda^4 + a_1 \lambda^3 + a_2 \lambda^2 + a_3 \lambda + a_4 = 0 \quad (3.14)$$

where

$$a_1 = 3d + \alpha + \alpha_1 + m_{22} \\ a_2 = (\alpha + d)(\alpha_1 + d) + d(\alpha + \alpha_1 + 2d + m_{22}) + m_{22}(\alpha + \alpha_1 + 2d) + m_{23}\varepsilon\delta + (1 - \varepsilon)\delta m_{24} \\ a_3 = d(\alpha + d)(\alpha_1 + d) + m_{22}[(\alpha + d)(\alpha_1 + d) + (\alpha + d)\alpha_1 + (\alpha_1 + d)d] + m_{24}[(1 - \varepsilon)\delta(\alpha_1 + 2d)] \\ \quad m_{24}\varepsilon\delta\alpha_1 + m_{23}(\alpha + d)\varepsilon\delta + m_{21}(1 - \varepsilon)\alpha\delta \\ a_4 = m_{22}d(\alpha + d)(\alpha_1 + d) + m_{23}d(\alpha + d)\varepsilon\delta + m_{24}d[\varepsilon\delta\alpha_1 + (1 - \varepsilon)\delta(\alpha_1 + d)] + m_{21}\alpha\varepsilon\delta\alpha_1 \\ \quad + m_{21}\alpha(1 - \varepsilon)\delta(\alpha_1 + d).$$

It can be seen easily that  $a_i > 0$  ( $i = 1, 2, 3, 4$ ) and  $a_1 a_2 - a_3 > 0$  under the sufficient condition on  $m_{21}$  as given above. Thus,  $E^*$  is locally asymptotically stable if it can be shown that  $a_3(a_1 a_2 - a_3) - a_1^2 a_4 > 0$  is satisfied.

Now to show that  $E^*$  is globally asymptotically stable, we first establish a lemma.

**Lemma 1.** *The region*

$$\Omega = \{(N, I, P, A); 0 < N(t) \leq \bar{N}; 0 \leq I(t) + P(t) \leq \bar{I}; 0 \leq A(t) \leq \bar{A}\} \quad (3.15)$$

is a region of attraction starting in the region  $[0 \leq I, P, A \text{ and } N \geq I + P + A]$ , where  $\bar{N} = \frac{Q_1 + Q_2}{d} + \kappa$ ,  $\bar{I} = \frac{Q_1 + Q_2}{d} \left[ \frac{[\beta_1 - (d + \zeta)] + \sqrt{[\beta_1 - (d + \zeta)]^2 + \frac{4Q_2\beta_2 d}{Q_1 + Q_2}}}{2\beta_2} \right]$ ,  $\bar{A} = \frac{\xi \bar{I}}{(\alpha + d)}$ ,  $\beta_1 = \max(\beta c, \beta' c)$ ,  $\beta_2 = \min(\beta c, \beta' c)$ ,  $\zeta = \min[\delta(1 - \varepsilon), \alpha_1]$ ,  $\xi = \max[\delta(1 - \varepsilon), \alpha_1]$  and  $\kappa > 0$  is an arbitrarily small constant.

The expression for  $\bar{I}$  has two values: one positive and one negative. We use only the positive value for  $\bar{I}$  (See [Appendix A](#) for detailed proof.).

**Theorem 1.** *If the endemic equilibrium  $E^*$  exists, then it is globally asymptotically stable provided the following sufficient conditions are satisfied in  $\Omega$  (see [Appendix B](#) for proof),*

$$9\alpha^2(1 - \varepsilon)\delta m_{24} < 4d^2(\alpha + d) \left[ m_{24} + \frac{Q_2}{\bar{I}} \right] \quad (3.16)$$

$$3\alpha_1^2 \varepsilon m_{24} < 2(1 - \varepsilon)(\alpha_1 + d)(\alpha + d)(m_{24} - \beta' c). \quad (3.17)$$

**Remark 1.** It is noted that system (2.5)–(2.8) is bounded by the same system when  $\alpha = 0$  and  $\alpha_1 = 0$ , which implies that the solution of (2.5)–(2.8) is bounded by the solution of the latter. If  $\alpha = 0$  and  $\alpha_1 = 0$ , the system is globally stable without any condition. Hence we speculate that the nontrivial equilibrium of system (2.5)–(2.8) may be globally stable as given in the theorem.

This theorem implies that under certain conditions, the disease would persist and remain endemic in the population. Moreover, the direct inflow of infectives makes the disease more endemic.

In the following sections, we consider the particular cases of the model (2.5)–(2.8) under different epidemiological situations to study the model dynamics.

#### 4. Model without inflow of HIV infectives including interaction with pre-AIDS individuals ( $\beta' \neq 0$ and $Q_2 = 0$ )

In this case, we consider the situation when all the immigrants are susceptible without direct inflow of HIV infectives and the individuals in pre-AIDS class also interact sexually but their interaction rate is very low i.e.  $\beta' \ll \beta$ . Though this may not be the case in developed countries owing to high level of awareness but certainly this may be true to some extent in underdeveloped or poor countries where some people may not even know that they are infected to this level by the disease.

In this case, since there is no direct inflow of infectives, the model exhibits two non-negative equilibria namely, the disease-free equilibrium and the endemic equilibrium.

(i)  $E_0 = (Q_1/d, 0, 0, 0)$ , the disease free.

It is found that the equilibrium  $E_0$  is locally asymptotically stable if,  $[\beta' c \varepsilon \delta + \beta c(\alpha_1 + d) < (\alpha_1 + d)(\delta + d)]$  holds.

We define the basic reproduction number  $R_0$  as, [12,4]

$$R_0 = \frac{[\beta' c \varepsilon \delta + \beta c(\alpha_1 + d)]}{(\alpha_1 + d)(\delta + d)}. \quad (4.1)$$

Using  $R_0$ , we can get the following condition for the stability of  $E_0$ .

**Lemma 2.** *The disease-free equilibrium  $E_0$  is locally asymptotically stable if  $R_0 < 1$  and unstable if  $R_0 > 1$ .*

From [Lemma 2](#), when the basic reproduction number  $R_0 > 1$ , the trivial equilibrium becomes unstable. Under this condition, there exists a nontrivial equilibrium i.e. endemic equilibrium, denoted by  $\hat{E}$  exists and the infection is maintained in the population.

(ii)  $\hat{E} = (\hat{N}, \hat{I}, \hat{P}, \hat{A})$ , the endemic equilibrium where,

$$\hat{N} = \left( \frac{Q_1 R_0 (1 + \pi + \pi \tau)}{\alpha (R_0 - 1) + R_0 (1 + \pi + \pi \tau) d} \right), \quad \hat{I} = \tau \pi \left( \frac{Q_1 (R_0 - 1)}{\alpha (R_0 - 1) + R_0 (1 + \pi + \pi \tau) d} \right)$$

$$\hat{P} = \pi \left( \frac{Q_1 (R_0 - 1)}{\alpha (R_0 - 1) + R_0 (1 + \pi + \pi \tau) d} \right), \quad \hat{A} = \left( \frac{Q_1 (R_0 - 1)}{\alpha (R_0 - 1) + R_0 (1 + \pi + \pi \tau) d} \right).$$

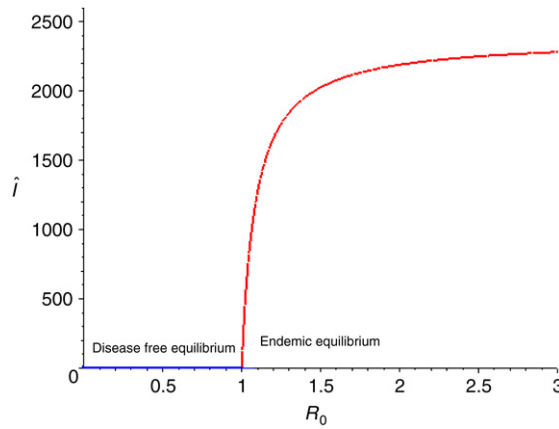


Fig. 2. Plot  $\hat{I}$  vs  $R_0$ .

And  $\pi$  and  $\tau$  are same as defined in Section 3.1. It is noted that  $\hat{E}$  is positive if  $R_0 > 1$ .

It is seen that  $\hat{I}$  increases as  $Q_1$  increases leading to increase in  $\hat{P}$  and  $\hat{A}$ . Further  $\hat{A}$  decreases as  $\alpha$  increases and  $\hat{P}$  increases as  $\alpha_1$  decreases under the condition  $R_0 > 1$ . It is also noted that when the disease remain endemic, the disease-induced deaths reduce the equilibrium population size from  $Q_1/d$  to  $\hat{N}$ . Also, since  $\hat{N} < N$ , it follows that  $I > \hat{I}$ ,  $P > \hat{P}$  and  $A > \hat{A}$ .

For System (2.5)–(2.8) when  $Q_2 = 0$ ,  $R_0 = \frac{[\beta'c\epsilon\delta + \beta c(\alpha_1 + d)]}{(\alpha_1 + d)(\delta + d)}$ . A transcritical bifurcation in the equilibria occurs at  $R_0 = 1$  (see Fig. 2).

If  $R_0 < 1$ , no biologically meaningful endemic equilibrium exists, and the disease-free equilibrium is an attractor. But if  $R_0 > 1$ , the endemic equilibrium exists and is an attractor, while the disease-free equilibrium is a saddle point [27,28]. Thus, there is a forward bifurcation because in the neighborhood of the bifurcation point, the endemic disease prevalence is an increasing function of  $R_0$ .

**Corollary 1.** The endemic equilibrium  $\hat{E}$  is locally asymptotically stable under the same conditions as in Section 3.2 for  $E^*$  when  $Q_2 = 0$ .

**Corollary 2.** The endemic equilibrium  $\hat{E}$  is also found to be globally asymptotically stable under the conditions (3.16)–(3.17) when  $Q_2 = 0$  in the region of attraction that can be easily obtained by putting  $Q_2 = 0$  in Lemma 1.

## 5. Model without inflow of HIV infectives and no interaction with pre-AIDS individuals ( $\beta' = 0$ and $Q_2 = 0$ )

In this case, we consider the situation when all the immigrants are susceptible without direct inflow of HIV infectives and the individuals in pre-AIDS class do not take part in sexual interaction as they may either be aware of their infection or sexually inactive. Thus, they may not contribute to spread the disease.

The model, in this case, again has two non-negative equilibria namely  $E_0(Q_1/d, 0, 0, 0)$ , the disease free, and  $\tilde{E}(\tilde{N}, \tilde{I}, \tilde{P}, \tilde{A})$  the endemic equilibrium which is positive if  $R_{01} > 1$ . The endemic equilibrium values  $(\tilde{N}, \tilde{I}, \tilde{P}, \tilde{A})$  are explicitly given as follows,

$$\begin{aligned} \tilde{N} &= \frac{R_{01}(1 + \pi + \pi\tau)Q_1}{\alpha(R_{01} - 1) + dR_{01}(1 + \pi + \pi\tau)}, & \tilde{I} &= \frac{\tau\pi Q_1(R_{01} - 1)}{\alpha(R_{01} - 1) + dR_{01}(1 + \pi + \pi\tau)}, \\ \tilde{P} &= \frac{\pi Q_1(R_{01} - 1)}{\alpha(R_{01} - 1) + dR_{01}(1 + \pi + \pi\tau)} & \text{and} & \quad \tilde{A} = \frac{Q_1(R_{01} - 1)}{\alpha(R_{01} - 1) + dR_{01}(1 + \pi + \pi\tau)}. \end{aligned}$$

As in Section 4, we find that the equilibrium  $E_0$  is locally asymptotically stable provided  $R_{01} < 1$  where  $R_{01}$  is the basic reproduction number given by,

$$R_{01} = \frac{\beta c}{\delta + d} \quad (5.1)$$

and under this condition the equilibrium  $\tilde{E}$  does not exist. However, if  $R_{01} > 1$  the equilibrium  $E_0$  is a saddle point which is stable in  $N$ - $P$ - $A$  manifold and unstable in  $I$ -direction. In such a case  $\tilde{E}$  exists and the infection is maintained in the population. It is observed here that when pre-AIDS individuals do not take part in sexual interaction, the basic reproduction number is significantly reduced. Note from Eq. (4.1) that as  $\beta' \rightarrow 0$ ,  $R_0 \rightarrow R_{01}$ . Thus, the endemicity of the infection is reduced when pre-AIDS individuals do not take part in sexual interaction as they are either aware of their infection or have become sexually inactive (see also Figs. 7 and 8). Therefore, the sexual interaction of pre-AIDS individuals makes the disease more endemic.

**Corollary 3.** The equilibrium  $\tilde{E}$  is locally asymptotically stable under the conditions given in Section 3.2 for  $E^*$  when  $Q_2 = 0$  and  $\beta' = 0$ .

**Corollary 4.** The endemic equilibrium  $\tilde{E}$  is also globally asymptotically stable under the conditions (3.16)–(3.17) when  $Q_2 = 0$  and  $\beta' = 0$  in the region of attraction  $\Omega_1$  stated as follows [17],

$$\Omega_1 = \{(N, I, P, A); 0 < N(t) \leq \bar{N}; 0 \leq I(t) \leq \bar{I}; 0 \leq P(t) \leq \bar{P}; 0 \leq A(t) \leq \bar{A}\}$$

where

$$\bar{N} = \frac{Q_1}{d}, \quad \bar{I} = \frac{Q_1}{d} \left\{ 1 - \frac{(\delta + d)}{\beta c} \right\}, \quad \bar{P} = \frac{\varepsilon \delta \bar{I}}{(\alpha_1 + d)} \quad \text{and} \quad \bar{A} = \frac{\delta \bar{I}}{(\alpha + d)} \left\{ \frac{\alpha_1 + d(1 - \varepsilon)}{\alpha_1 + d} \right\}.$$

**Remark 2.** By analyzing the epidemiological situations as discussed above, it may be concluded that if there is direct inflow of infectives, there exists only one endemic equilibrium so that disease is always persistent. However, if inflow of infected immigrants is restricted then the endemicity of disease is still higher when pre-AIDS individuals also interact sexually. But the endemicity is significantly reduced when pre-AIDS individuals do not interact sexually and inflow of infected individuals is restricted.

## 6. Numerical analysis and discussion

To study the dynamical behavior of the model numerically, the system (2.5)–(2.8) is integrated by fourth order Runge–Kutta method using the following set of parameter values with units taken in per year, [17,18]

$$Q_1 = 1000, \quad Q_2 = 1000, \quad \varepsilon = 0.6, \quad d = 0.02, \quad \alpha = 1, \\ \alpha_1 = 0.5, \quad \beta = 0.15, \quad \beta' = 0.05, \quad c = 10, \quad \delta = 0.4$$

with initial values

$$N(0) = 5000, \quad I(0) = 2000, \quad P(0) = 1000, \quad A(0) = 500.$$

The equilibrium values are computed as,

$$N^* = 9863, \quad I^* = 4706, \quad P^* = 2172, \quad A^* = 1803.$$

Eigen values corresponding to  $E^*$  are obtained as,

$$-0.1629590182 \pm i0.3088089658, \quad -1.297258108, \quad -1.002787998.$$

Since all the eigen values corresponding to  $E^*$  are negative or have negative real parts, therefore  $E^*$  is locally asymptotically stable.

The computer simulation is also performed for different initial starts in the following four cases,

1.  $N(0) = 15000, I(0) = 8000, P(0) = 2000$  and  $A(0) = 200$ ,
2.  $N(0) = 11000, I(0) = 8000, P(0) = 2000$  and  $A(0) = 500$ ,
3.  $N(0) = 8000, I(0) = 1500, P(0) = 4000$  and  $A(0) = 100$ ,
4.  $N(0) = 5000, I(0) = 300, P(0) = 1000$  and  $A(0) = 400$ .

The results of numerical simulation are displayed graphically in Figs. 3–12. In Figs. 3 and 4, the infective population and pre-AIDS population respectively is plotted against the susceptible population. We see from these figures that for any initial start, the solution curve tend to the equilibrium  $E^*$ . Hence, we infer that the system (2.5)–(2.8) may be globally stable about this endemic equilibrium point  $E^*$ . The global stability conditions are found to be satisfied for the above set of parameter values. It may be noted here that we do not observe any limit cycle phenomenon for the violation of the conditions (3.16)–(3.17). In Fig. 5, the variation of population with time is shown for different classes without immigration. It is seen that in the absence of immigration into community, the susceptible population decreases continuously which results in an increase in infective population first and then it decreases as all infectives will develop AIDS and will die out by disease-induced deaths. Fig. 6 shows the variation of population in all classes with immigration. It is found that susceptible population first increases with time and then reaches its equilibrium position. It is expected that due to immigration susceptible population increases continuously, therefore, infection becomes more endemic and always persist in the population. In Figs. 7 and 8, the variation of HIV infected population and that of AIDS patients population is shown for different rates of inflow of infectives and contact rates of pre-AIDS patients. It is clear that when the pre-AIDS patients do not take part in sexual interaction and the direct inflow of infectives is also restricted, the number of infectives decreases leading to decline in AIDS population. Fig. 9 shows the variation of infective population with rate of inflow of infectives  $Q_2$ . It is clear that if direct inflow of infected immigrants increases, the number of infectives increases which ultimately increases the prevalence of the disease. Thus, if the inflow of infected immigrants into the community is restricted, the spread of the disease may be slowed down. The effect of increasing the number of sexual partners on the infective population is shown in

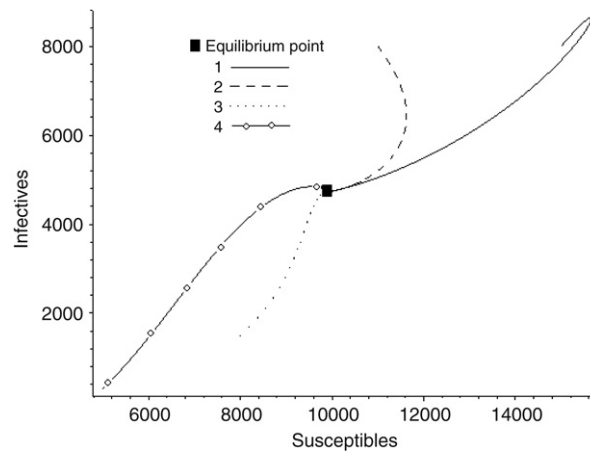


Fig. 3. Variation of infective population against susceptible population.

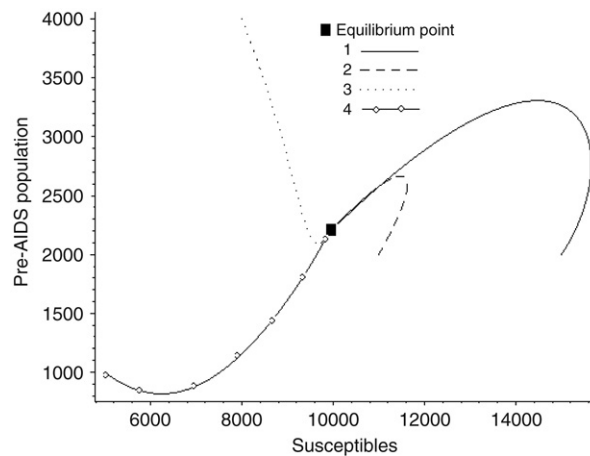


Fig. 4. Variation of pre-AIDS population against susceptible population.

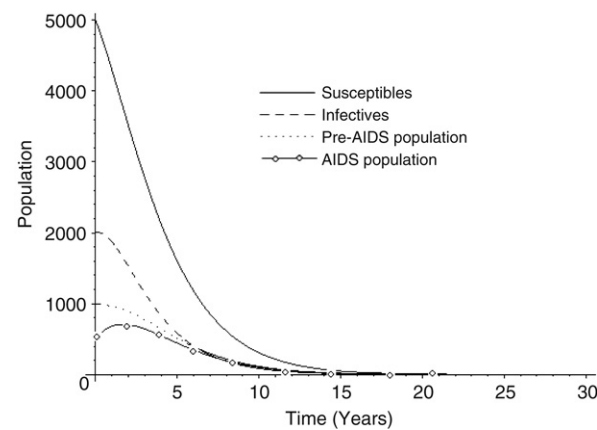


Fig. 5. Variation of population in different classes without immigration.

Fig. 10 and it is seen that if the number of sexual partners is higher the risk of infection increases. Thus, to reduce the spread of infection it is desired to keep the number of sexual partners at minimum or restrict the unsafe sexual interaction by way of using condoms etc. The role of  $\alpha_1$  i.e. the rate by which patients in pre-AIDS class proceed to develop full blown AIDS is plotted in Figs. 11 and 12. It is found that with increase in  $\alpha_1$ , the population in pre-AIDS class decreases whereas that of AIDS patients increases.

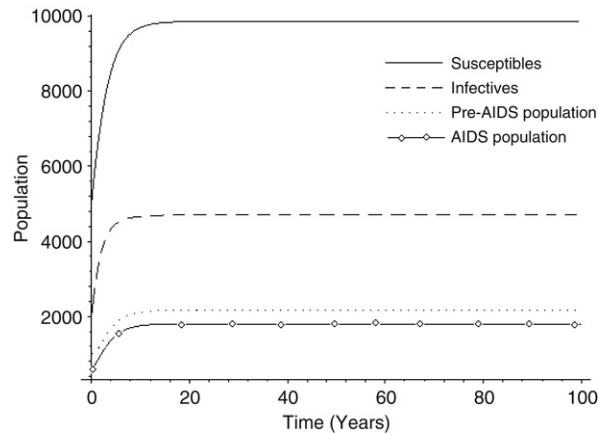


Fig. 6. Variation of population in different classes with immigration.

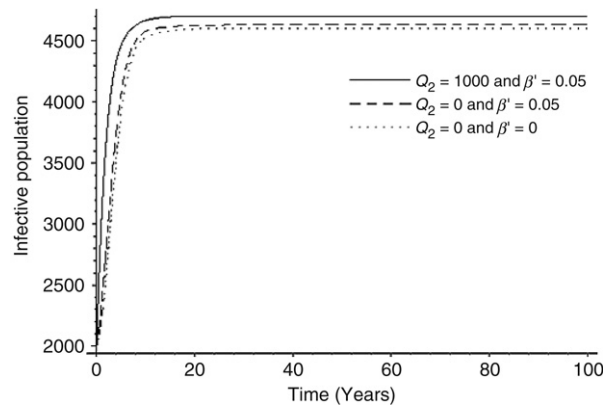


Fig. 7. Variation of infective population for different inflow rates of infective immigrants and contact rates  $\beta'$  of susceptibles with pre-AIDS individuals.

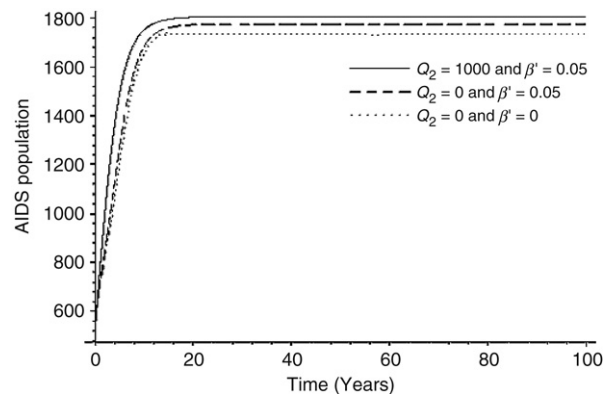


Fig. 8. Variation of AIDS population for different inflow rates of infective immigrants and contact rates  $\beta'$  of susceptibles with pre-AIDS individuals.

The models discussed above can also be extended in various ways. For example, if it is desired to incorporate the role of screening of infectives who are not aware of their infection, the role of contact tracing, the role of treatment given to incoming infectives and to assess the form of epidemic if the strategies of restricting the movement of infectives is combined with condom use, etc.

## 7. Conclusion

In this paper, a nonlinear mathematical model is proposed and analyzed to study the spread of HIV/AIDS with direct inflow of infectives in a population with variable size structure. It is assumed that susceptibles become infected via sexual

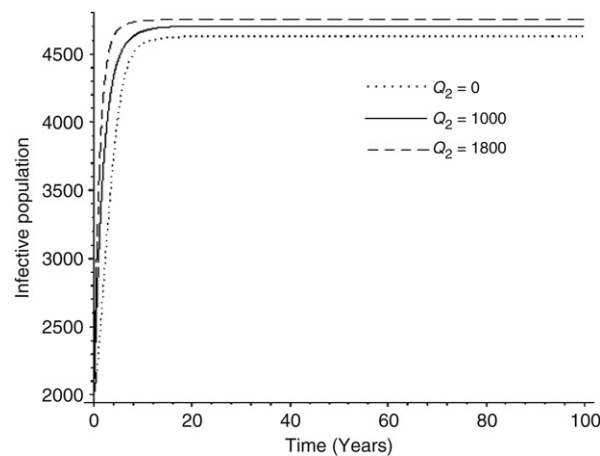


Fig. 9. Variation of infective population for different inflow rates of immigrated infectives.

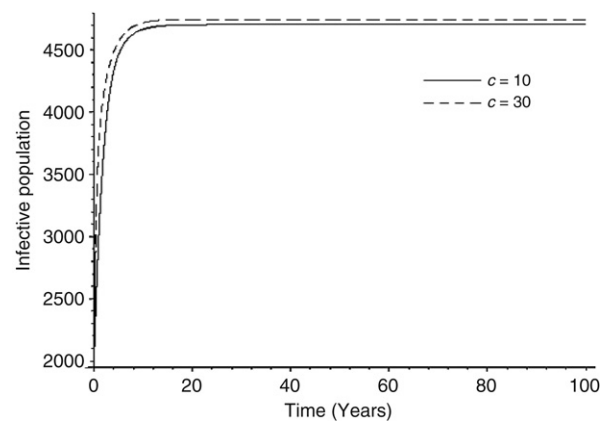


Fig. 10. Variation of infective population for different values of sexual partners  $c$ .

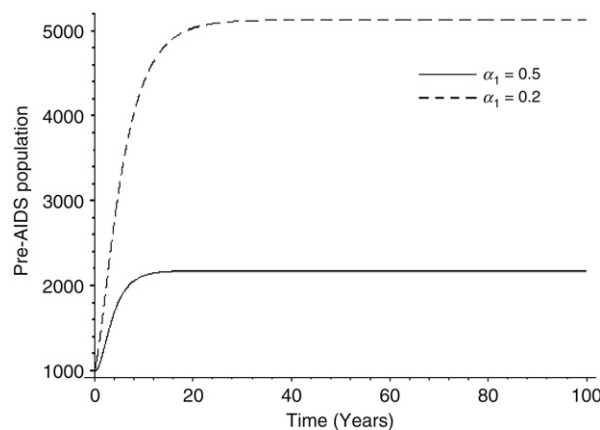


Fig. 11. Variation of pre-AIDS population for different values of movement rate  $\alpha_1$  of pre-AIDS individuals.

contacts with infectives and all infectives ultimately develop AIDS. The model is analyzed using stability theory of differential equations and numerical simulation. Two particular epidemiological situations have also been discussed. Some inferences have been drawn regarding the spread of the disease by way of establishing local and global stability results. It is found that if the direct inflow of the infectives is allowed in the community, the disease is always persistent. If inflow of infected immigrants is restricted then the endemicity of the disease is higher when pre-AIDS individuals also interact sexually. But the endemicity is significantly reduced if direct inflow of infectives is restricted and pre-AIDS individuals also do not take

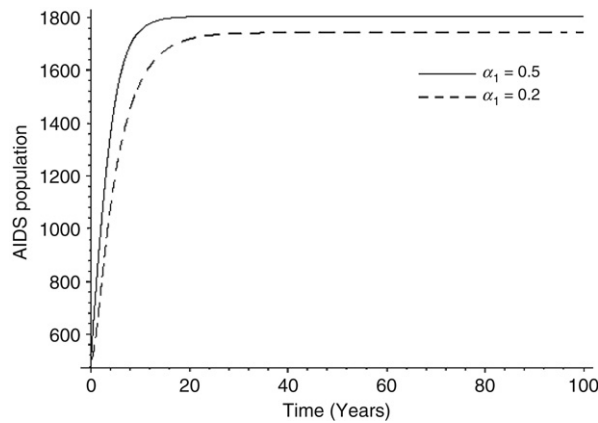


Fig. 12. Variation of AIDS population for different values of movement rate  $\alpha_1$  of pre-AIDS individuals.

part in sexual interactions. Thus the spread of infection can be slowed down if direct inflow of infectives is restricted into the population.

It is also noted that the increase in the number of sexual partners further reduces the total population by way of spreading the disease. Thus in order to reduce the spread of the disease, the number of sexual partners should be restricted and unsafe sexual interaction should be avoided with an infective.

### Acknowledgements

Authors are thankful to the anonymous referees whose comments and constructive suggestions helped them in finalizing the paper.

### Appendix A. Proof of Lemma 1

Since

$$\begin{aligned}\frac{dN}{dt} &= Q_1 + Q_2 - dN - \alpha A \\ &\leq Q_1 + Q_2 - dN.\end{aligned}$$

Thus,  $\lim_{t \rightarrow \infty} \sup N(t) \leq \frac{Q_1 + Q_2}{d} + \kappa$ , where  $\kappa > 0$  is an arbitrarily small constant

$$\begin{aligned}\frac{dI}{dt} + \frac{dP}{dt} &= Q_2 + \frac{\beta c(N - I - P - A)I}{N} + \frac{\beta' c(N - I - P - A)P}{N} - d(I + P) - \delta(1 - \varepsilon)I - \alpha_1 P \\ \frac{d}{dt}(I + P) &= Q_2 + \beta cI + \beta' cP - d(I + P) - (\beta cI + \beta' cP) \left( \frac{I + P + A}{N} \right) - [\delta(1 - \varepsilon)I + \alpha_1 P] \\ &\leq Q_2 + [\beta_1 - d](I + P) - \frac{\beta_2(I + P)(I + P)}{\bar{N}} - \zeta(I + P) \\ &= Q_2 + [\beta_1 - (d + \zeta)](I + P) - \frac{\beta_2 d(I + P)^2}{Q_1 + Q_2}.\end{aligned}$$

Here right-hand side is a quadratic equation in  $(I + P)$

$$\frac{\beta_2 d(I + P)^2}{Q_1 + Q_2} - [\beta_1 - (d + \zeta)](I + P) - Q_2 = 0.$$

Therefore,

$$(I + P) = \frac{Q_1 + Q_2}{d} \left[ \frac{[\beta_1 - (d + \zeta)] \pm \sqrt{[\beta_1 - (d + \zeta)]^2 + \frac{4Q_2\beta_2 d}{Q_1 + Q_2}}}{2\beta_2} \right].$$

Here two roots of  $(I + P)$  exist but the positive value of  $(I + P)$  is given as,

$$\text{where } \bar{I} = \frac{Q_1 + Q_2}{d} \left[ \frac{[\beta_1 - (d + \zeta)] + \sqrt{[\beta_1 - (d + \zeta)]^2 + \frac{4Q_2\beta_2 d}{Q_1 + Q_2}}}{2\beta_2} \right].$$

Thus,  $\lim_{t \rightarrow \infty} \sup[I(t) + P(t)] \leq \bar{I}$ .

It may be noted that negative value of  $\bar{I}$  is biologically irrelevant, therefore, only the positive value should be used. Similarly

$$\begin{aligned}\frac{dA}{dt} &= (1 - \varepsilon)\delta I + \alpha_1 P - (\alpha + d)A \\ &\leq \xi(I + P) - (\alpha + d)A \\ &\leq \xi\bar{I} - (\alpha + d)A.\end{aligned}$$

Then  $\lim_{t \rightarrow \infty} \sup A(t) \leq \bar{A}$

where  $\bar{A} = \frac{\xi\bar{I}}{(\alpha+d)}$  proving the lemma.

## Appendix B. Proof of Theorem 1

Consider the following positive definite function about  $E^*$ ,

$$V = \frac{1}{2}(N - N^*)^2 + k_1 \left( I - I^* - I^* \ln \frac{I}{I^*} \right) + \frac{1}{2}k_2 (P - P^*)^2 + \frac{1}{2}k_3 (A - A^*)^2. \quad (\text{B.1})$$

Differentiating  $V$  with respect to  $t$ , we get

$$\frac{dV}{dt} = (N - N^*) \frac{dN}{dt} + k_1 \left( \frac{I - I^*}{I} \right) \frac{dI}{dt} + k_2 (P - P^*) \frac{dP}{dt} + k_3 (A - A^*) \frac{dA}{dt}.$$

Using Eqs. (2.5)–(2.8) and simplifying, we get

$$\begin{aligned}\frac{dV}{dt} &= -d(N - N^*)^2 + k_1 \left[ \frac{\beta c}{N^*} + \frac{\beta' c P^*}{N^* I^*} \right] \left( \frac{I + P + A}{N} \right) (N - N^*) (I - I^*) - \alpha (N - N^*) (A - A^*) \\ &\quad - k_1 \left[ \frac{\beta c}{N^*} + \frac{\beta' c P^*}{N^* I^*} + \frac{Q_2}{I I^*} + \frac{\beta' c P}{I^* I} \left( \frac{N - I - P - A}{N} \right) \right] (I - I^*)^2 - k_2 (\alpha_1 + d) (P - P^*)^2 \\ &\quad + \left[ k_2 \varepsilon \delta - k_1 \left\{ \frac{\beta c}{N^*} + \frac{\beta' c P^*}{N^* I^*} - \frac{\beta' c}{I^*} \left( \frac{N - I - P - A}{N} \right) \right\} \right] (I - I^*) (P - P^*) \\ &\quad - k_3 (\alpha + d) (A - A^*)^2 + \left[ k_3 (1 - \varepsilon) \delta - k_1 \left( \frac{\beta c}{N^*} + \frac{\beta' c P^*}{N^* I^*} \right) \right] (I - I^*) (A - A^*) + k_3 \alpha_1 (P - P^*) (A - A^*).\end{aligned}$$

After little algebraic calculations, it can be further written as the sum of the quadratics.

Now for  $dV/dt$  to be negative definite, the following conditions must be satisfied,

$$k_1 \left[ \frac{m_{24}}{I^*} \left( \frac{I + P + A}{N} \right) \right]^2 < \frac{2d}{3} \left[ \frac{m_{24}}{I^*} + \frac{Q_2}{I I^*} + \frac{\beta' c P}{I^* I} \left( \frac{N - I - P - A}{N} \right) \right] \quad (\text{B.2})$$

$$\alpha^2 < \frac{2d}{3} k_3 (\alpha + d) \quad (\text{B.3})$$

$$\left[ k_2 \varepsilon \delta - k_1 \left\{ \frac{m_{24}}{I^*} - \frac{\beta' c}{I^*} \left( \frac{N - I - P - A}{N} \right) \right\} \right]^2 < \frac{2(\alpha_1 + d)}{3} \left[ \frac{m_{24}}{I^*} + \frac{Q_2}{I I^*} + \frac{\beta' c P}{I^* I} \left( \frac{N - I - P - A}{N} \right) \right] k_1 k_2 \quad (\text{B.4})$$

$$\left[ k_3 (1 - \varepsilon) \delta - k_1 \frac{m_{24}}{I^*} \right]^2 < \frac{4(\alpha + d)}{9} \left[ \frac{m_{24}}{I^*} + \frac{Q_2}{I I^*} + \frac{\beta' c P}{I^* I} \left( \frac{N - I - P - A}{N} \right) \right] k_1 k_3 \quad (\text{B.5})$$

$$k_3 \alpha_1^2 < \frac{2}{3} k_2 (\alpha + d) (\alpha_1 + d). \quad (\text{B.6})$$

After maximizing the LHS and minimizing the RHS of the above, the stability conditions can be obtained as follows,

$$9\alpha^2 (1 - \varepsilon) \delta m_{24} < 4d^2 (\alpha + d) \left[ m_{24} + \frac{Q_2}{I} \right] \quad (\text{B.7})$$

$$3\alpha_1^2 \varepsilon m_{24} < 2(1 - \varepsilon)(\alpha_1 + d)(\alpha + d)(m_{24} - \beta' c) \quad (\text{B.8})$$

where the constant  $k_1 > 0$  can be chosen such that

$$\frac{\alpha^2 \delta (1 - \varepsilon) I^*}{d(\alpha + d) m_{24}} < k_1 < \frac{d \left[ m_{24} + \frac{Q_2}{I} \right] I^*}{2(m_{24})^2} \quad \text{and} \quad k_2 = \frac{k_1}{\varepsilon \delta I^*} [m_{24} - \beta' c], \quad k_3 = \frac{k_1 m_{24}}{(1 - \varepsilon) \delta I^*}.$$

Under conditions (B.7) and (B.8),  $dV/dt$  will be negative definite showing that  $V$  is a Lyapunov function with respect to  $E^*$  whose domain contains  $\Omega$ .

## References

- [1] C. Piccolo III, L. Billings, The effect of vaccinations in an immigrant model, *Math. Comp. Mod.* 42 (2005) 291–299.
- [2] E. Shim, A note on epidemic models with infective immigrants and vaccination, *Math. Bios. Engg.* 3 (2006) 557–566.
- [3] L.R. Krilov, Emerging infectious disease issues in international adoptions: SARS, avian influenza and measles, *Curr. Opin. Infect. Dis.* 17 (2004) 391–395.
- [4] Z. Mukandavire, W. Garira, C. Chiyaka, Asymptotic properties of an HIV/AIDS model with a time delay, *J. Math. Anal. Appl.* 330 (2007) 916–933.
- [5] R.M. Anderson, G.F. Medley, R.M. May, A.M. Johnson, A preliminary study of the transmission dynamics of the human immunodeficiency virus (HIV), the causative agent of AIDS, *IMA J. Math. Appl. Med. Biol.* 3 (1986) 229–263.
- [6] R.M. Anderson, The role of mathematical models in the study of HIV transmission and the epidemiology of AIDS, *J. AIDS* 1 (1988) 241–256.
- [7] R.M. May, R.M. Anderson, Transmission dynamics of HIV infection, *Nature* 326 (1987) 137–142.
- [8] J. Bongaarts, A model for the spread of HIV infection and the demographic impact of AIDS, *Statist. Med.* 8 (1989) 103–120.
- [9] S. Busenberg, K. Cooke, H. Ying-Hen, A model for HIV in Asia, *Math. Bios.* 128 (1995) 185–210.
- [10] S. Busenberg, P. van den Driessche, Analysis of a disease transmission model in a population with varying size, *J. Math. Biol.* 28 (1990) 257–270.
- [11] H. De Arazoza, R. Lounes, A nonlinear model for a sexually transmitted disease with contact tracing, *IMA J. Math. Appl. Med. Biol.* 19 (2002) 221–234.
- [12] M. Doyle, D. Greenhalgh, S. Blythe, Equilibrium analysis of a mathematical model for the spread of AIDS in a two sex population with mixing constraints, *J. Biol. Syst.* 6 (1998) 159–185.
- [13] J. Gani, S. Yakowitz, M. Blount, The spread and quarantine of HIV infection in a prison system, *SIAM J. Appl. Math.* 57 (1997) 1510–1530.
- [14] G.P. Garnett, R.M. Anderson, Sexually transmitted diseases and sexual behavior: insights from mathematical models, *J. Infect. Diseases* 174 (1996) S150–S161.
- [15] D. Greenhalgh, M. Doyle, F. Lewis, A mathematical treatment of AIDS and condom use, *IMA J. Math. Appl. Med. Biol.* 18 (2001) 225–262.
- [16] J.S. Koopman, J.A. Jacquez, G.W. Welch, C.P. Simon, B. Foxman, S.M. Pollock, D. Brath-Jones, A.L. Adams, K. Lange, The role of early HIV infection in the spread of HIV through populations, *J. AIDS* 14 (1997) 249–258.
- [17] R. Naresh, A. Tripathi, S. Omar, Modelling the spread of AIDS epidemic with vertical transmission, *Appl. Math. Comput.* 178 (2006) 262–272.
- [18] A. Tripathi, R. Naresh, D. Sharma, Modelling the screening of unaware infectives on the spread of HIV infection, *Appl. Math. Comput.* 184 (2007) 1053–1068.
- [19] B.D. Aggarwala, On two ODE models for HIV/AIDS development in Canada and a logistic SEIR model, *Far East J. Appl. Math.* 6 (2002) 25–70.
- [20] A.S.R. Srinivasa Rao, Mathematical modeling of AIDS epidemic in India, *Current Sci.* 84 (2003) 1192–1197.
- [21] M. Zhien, J. Liu, Stability analysis for differential infectivity epidemic models, *Nonlinear Anal. RWA* 4 (2003) 841–856.
- [22] F. Brauer, P. van den Driessche, Models for transmission of disease with immigration of infectives, *Math. Bios.* 171 (1995) 143–154.
- [23] C.M. Kribs-Zaleta, M. Lee, C. Roman, S. Wiley, C.M. Hernandez-Suarez, The effect of the HIV/AIDS epidemic on Africa's truck drivers, *Math. Bios. Engg.* 2 (2005) 771–788.
- [24] Y.H. Hsieh, C.H. Chen, Modelling the social dynamics of a sex industry: Its implications for spread of HIV/AIDS, *Bull. Math. Biol.* 66 (2004) 143–166.
- [25] J.M. Hyman, J. Li, E.A. Stanley, The differential infectivity and staged progression models for the transmission of HIV, *Math. Bios.* 155 (1999) 77–109.
- [26] X. Lin, H.W. Hethcote, P. van den Driessche, An epidemiological model for HIV/AIDS with proportional recruitment, *Math. Bios.* 118 (1993) 181–195.
- [27] R.M. Anderson, R.M. May, *Infectious Diseases of Humans*, in: Dynamics and Control, Oxford University Press, Oxford, 1991.
- [28] A. Korobeinikov, G.C. Wake, Lyapunov functions and global stability for SIR, SIRS, and SIS epidemiological models, *Appl. Math. Lett.* 15 (2002) 955–960.